One-dimensional simulation of geysering driven by entrapped-air release through water-filled vertical shafts using a decoupled two-fluid mixed-flow framework

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Abstract

Geysering in stormwater and combined-sewer systems occurs when entrapped air pockets are released through vertical shafts and expel the overlying water column above ground. System-scale prediction relies on reduced one-dimensional models, but most existing mixed-flow formulations prescribe the air phase through boundary data or lumped pocket laws, so the coupled air-pocket–water-column dynamics that select between benign venting and geysering are not part of the computed state. This paper applies a one-dimensional decoupled two-fluid cut-cell finite-volume framework, developed in a companion methods paper, to laboratory geysering experiments. The framework retains gas mass and gas momentum as conserved variables in stratified reaches, treats moving pressurized–stratified fronts as Regime-Transition Riemann problems, and represents the vertical shaft as the same one-dimensional two-fluid system rotated to the vertical, coupled to the horizontal conduit through a junction with local-loss closures. All simulations start from the experimental initial conditions and use one frozen solver

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configuration, without prescribed release rates, scripted geyser heights, or per-case fitting. The model is first evaluated against published ventilation-shaft experiments in which a pressurized air pocket is released toward a partially filled tower, with a wide-tower configuration in the non-geysering branch and a narrow-tower configuration in the geysering branch. The correct branch is selected in both configurations. In the non-geysering branch, the computed pressure-head plateau, terminal venting decay, air–water interface climb velocity, and maximum free-surface rise agree with the measurements. In the geysering branch, the pre-eruption stand of the tower column, the pressure plateau, the eruption to the shaft rim observed in every experimental repetition, the interface and free-surface climb velocities, and the sharp head collapse at pocket venting are reproduced, with the gas-phase event times uniformly early by about half a non-dimensional time unit. The model is then evaluated as a regime classifier against published horizontal-pipe/vertical-riser experiments, using the same frozen configuration under the fully coupled junction treatment: it reproduces both end members of the published two-parameter geysering criterion—the narrow-riser eruption with its pocket-compression pressure surge, and the benign venting of every wide-riser and small-pocket configuration—and the measured branch differentiation of the pipe-end pressure peak. Its classification errors over the experimental parameter sweep are strictly one-sided: near the regime boundary it under-predicts eruption, driving the water column high into the riser without expelling it, and it produces no false geyser. The conservative bias is traced to a single identified closure—the volume-neutral gas–water exchange at the junction mouth, whose demands in the two branches conflict—together

with one-dimensional limitations in junction transmission and annular film entrainment. A campaign against prototype-motivated junction-chamber experiments will complete the evaluation in a subsequent revision.
Keywords: geysering, entrapped air pocket, stormwater system, vertical shaft, two-fluid model, transient mixed flow

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1. Introduction

Geysering denotes the violent expulsion of air–water mixtures through manholes, dropshafts, or ventilation shafts of stormwater and combined-sewer systems during rapid filling events. Field observations show mixtures rising tens of metres above ground, damaging manhole covers and endangering the public. Early explanations based on pure inertial oscillations of the water column were contradicted by field pressure records showing that the piezometric head near erupting shafts never approaches the ground surface [1]; the driving mechanism is instead the release of large entrapped air pockets through partially or fully water-filled vertical shafts [2, 1]. The laboratory experiments of Vasconcelos and Wright [2] established this air-release mechanism in a controlled single-shaft apparatus and remain the primary benchmark for geysering models. Subsequent experimental programmes examined the release of entrapped air from a horizontal pipe into a vertical shaft [3], water displacement and geysering in larger-scale shafts [4], stormwater geysers above junction chambers modelled on a prototype dropshaft [5], geysers through covered manholes [6], and mitigation measures for storm geysers [7]. The physical core of the event—a large gas pocket rising and expanding in a vertical water column until the expansion becomes violent—has also been isolated in the multiphase-flow literature on Taylor bubble expansion [8].

Numerical prediction of geysering spans two extremes. Three-dimensional volume-of-fluid simulations reproduce individual laboratory events in detail [10], but their cost restricts them to single shafts and short time windows, so they cannot be used for network-scale screening of hundreds of shafts under many storm scenarios. At the other extreme, one-dimensional transient mixed-flow models are fast enough for system-scale simulation, but most existing formulations treat the air phase incompletely. Preissmann-slot and two-component-pressure formulations follow the pressurization of the liquid phase [11], while entrapped air, when represented at all, enters through lumped pocket laws, prescribed release rates, or boundary conditions. Rigid-column two-phase models of the type used by Vasconcelos and Wright [2] to interpret their own experiments prescribe the topology of the air pocket and the water column in advance, which limits them to geometries where that topology is known. Because the selection between benign venting and geysering depends on the coupled dynamics of pocket compression, air intrusion into the shaft, and counter-current air–water exchange at the shaft mouth, a predictive one-dimensional model must carry the gas phase as part of the computed state rather than as a prescribed forcing.

A companion methods paper [12] developed a one-dimensional conservative finite-volume framework for transient closed-conduit mixed flows in which the stratified reaches are described by a decoupled two-fluid system that retains gas mass and gas momentum as conserved variables, liquid-full reaches use an elastic-pipe water-hammer closure, and moving pressurized–stratified fronts are treated as Regime-Transition Riemann problems embedded in a conservative cut-cell front-tracking update. In that framework, air-pocket compression, gas transport, and interfacial-instability growth are outcomes of the resolved gas–liquid dynamics. The methods paper verified the framework on interfacial-wave, slug-formation, air-pocket pressure, and integrated mixed-flow benchmarks, and included a schematic one-dimensional geysering-onset test; it did not, however, attempt a quantitative comparison against geysering experiments.

The present paper provides that application-level assessment. The framework of [12] is applied, unchanged in its formulation, to laboratory geysering experiments in which a vertical shaft is coupled to a horizontal conduit and an entrapped air pocket is released toward the shaft. The vertical shaft is discretized as the same one-dimensional two-fluid system rotated to the vertical and coupled to the horizontal pipe through a junction closure with physically motivated local losses. All regime selection—whether the free surface reaches the shaft rim before the rising air–water interface catches it—emerges from the computed state.

The evaluation is organized as test campaigns against published experiments, each contributing a different kind of evidence. The first campaign is the water-filled ventilation-shaft experiment of Vasconcelos and Wright [2], from which two configurations are selected to span the two observed regimes: a large-tower case in the non-geysering branch and a small-tower case in the geysering branch, testing the time-resolved pressure and interface histories. The second campaign is the horizontal-pipe/vertical-riser experiment of Cong, Chan and Lee [3], whose systematic parameter study condensed the geysering outcome into an explicit two-parameter criterion; it tests the model as a regime classifier across the high-speed-camera series and a 63-configuration sweep of the experimental parameter ranges. All computations use a frozen solver configuration and start from the experimental initial conditions, without prescribed release rates, scripted geyser heights, or per-case parameter fitting. A third campaign against the prototype-motivated junction-chamber experiments of Liu, Shao and Zhu [5] tests branch selection with a downstream tail boundary, eruption height against peak pressure, and the transient mass-oscillation model of their Series C cases; other experimental studies [4, 6, 7, 9] serve as corroborating references.

The remainder of the paper is organized as follows. Section 2 summarizes the decoupled two-fluid framework and describes the vertical-shaft branch and junction closures used for geysering configurations. Section 3 presents the test cases and the comparison with the experiments. Section 4 discusses the capabilities and limitations identified by the comparison, and Section 5 concludes.

2. Model framework

This section summarizes the one-dimensional framework of the companion methods paper [12] to the extent needed to interpret the geysering computations, and then describes the vertical-shaft branch, junction closures, and trapped-pocket treatment used in the geysering configurations. The derivations, the Regime-Transition Riemann solver, and the conservative cut-cell front-tracking algorithm are given in [12] and are not repeated here.

2.1. *Decoupled two-fluid mixed-flow formulation*

The framework describes a closed conduit as a union of liquid-full (pressurized) and stratified gas–liquid reaches separated by moving regime fronts. In stratified reaches the flow is governed by the decoupled two-fluid balance law of [12],

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{S}, \tag{1}$$

with conservative state, flux, and regular source vectors

$$\mathbf{U} = \begin{bmatrix} A_g \rho_g \\ \rho_g Q_g \\ A_l \\ Q_l \end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix} \rho_g Q_g \\ \rho_g Q_g^2/A_g + P_g A_g \\ Q_l \\ Q_l^2/A_l + \frac{1}{2}\Lambda_d A_l^2 \end{bmatrix}, \qquad \mathbf{S} = \begin{bmatrix} 0 \\ P_g \dfrac{\partial A_g}{\partial x} - A_g \rho_g g \cos\theta \dfrac{\partial h_l}{\partial x} + \mathcal{S}_g \\ 0 \\ \mathcal{S}_d \end{bmatrix}, \tag{2}$$

where A_g and A_l are the gas and liquid cross-sectional areas, ρ_g the gas density, Q_g and Q_l the gas and liquid volumetric flow rates, P_g the gas pressure, h_l the liquid depth, θ the pipe inclination, and $\mathcal{S}_g$, $\mathcal{S}_d$ the regular gas- and liquid-momentum sources containing axial gravity, wall friction, and interfacial shear; the shear closures follow standard stratified-flow correlations [13, 14]. Gas mass and gas momentum are conserved variables, so gas transport, gas inertia, gas compressibility, and shear-driven interfacial growth are part of the computed state. The restoring coefficient in the liquid momentum flux is

$$\Lambda_d = \underbrace{\frac{2gH_g}{A_l}}_{\text{gas pressure}} + \underbrace{\frac{\rho_l - \rho_g}{\rho_l} g\zeta}_{\text{hydrostatic–buoyancy}} - \underbrace{\frac{\rho_g (u_g - u_l)^2}{\rho_l A_g}}_{\text{IKH slip}}, \tag{3}$$

which combines the gas-pressure, hydrostatic–buoyancy, and inviscid Kelvin–Helmholtz (IKH) slip contributions; here H_g is the gas pressure head, u_g

and u_l the phase velocities, and $\zeta = \cos\theta/[D\sin(\gamma/2)]$ a circular-section geometric factor with D the pipe diameter and γ the liquid central angle. The eigenvalues of the decoupled liquid flux Jacobian,

$$\lambda^{\pm} = u_l \pm \sqrt{\Lambda_d A_l}, \tag{4}$$

show that the liquid subsystem is strictly hyperbolic for $\Lambda_d > 0$ and reaches the neutral interfacial-instability (IKH) threshold at $\Lambda_d = 0$; beyond it, interfacial waves grow and liquid bridging can occur.

In liquid-full reaches the gas phase is absent and the pressurized branch is governed by the two-variable system

$$\frac{\partial A_l}{\partial t} + \frac{\partial Q_l}{\partial x} = 0, \qquad \frac{\partial Q_l}{\partial t} + \frac{\partial}{\partial x}\left(\frac{Q_l^2}{A_l} + gA_f h_s\right) = \mathcal{S}_p, \tag{5}$$

closed by the elastic-pipe relation

$$P - P_{\text{ref}} = \rho_l a^2 \frac{A_l - A_f}{A_f}, \qquad h_s = \frac{P - P_{\text{ref}}}{\rho_l g}, \tag{6}$$

with a the (numerical) pressure-wave speed, A_f the undeformed pipe area, h_s the piezometric head, and $\mathcal{S}_p$ the regular pressurized-branch source; this branch recovers the water-hammer limit. In limiting cases the same area-variable form of Eqs. (1)–(6) recovers water hammer, open-channel gravity flow (with $\Lambda_d \to g\cos\theta/B_l$, where B_l is the free-surface top width), interfacial-instability dynamics, and the near-full mixed-flow balance [12].

Moving pressurized–stratified fronts are internal boundaries between the two-variable system (5) and the four-variable system (1). Their local closure and the conservative front-tracking update are summarized in Sections 2.2–2.3, so that the computations reported here can be interpreted and reproduced from the present paper alone; the derivations, the construction of the active-set solver, and the verification of the scheme on canonical benchmarks are given in [12].

Two properties of this formulation matter for geysering. First, because gas mass is conserved per computational region, entrapped pockets are identified as connected gas regions bounded by liquid seals, and their pressure follows from an isothermal equation of state $P_g = m_g RT/V_g$ evaluated with the current pocket mass m_g and volume V_g; no polytropic pocket law with a prescribed volume history is imposed. Second, because the stratified branch carries gas momentum, air intrusion into a shaft and counter-current air–water exchange at the shaft mouth are resolved rather than prescribed.

2.2. Regime-Transition Riemann closure at moving fronts

At a moving pressurized–stratified front $x = x_\Gamma(t)$ with speed $w_\Gamma = \dot{x}_\Gamma$, the local coordinate is oriented so that the pressurized branch lies on the left and the stratified branch on the right; the opposite case follows by reflection. Only liquid mass and liquid momentum are common to both branches, and the ideal front has zero thickness and stores neither. Integrating the mixed-regime liquid balance over a control volume shrinking onto the front therefore gives the moving Rankine–Hugoniot conditions

$$\begin{gathered} Q_{p,\Gamma} - Q_{f,\Gamma} = w_\Gamma \left(A_{p,\Gamma} - A_{l,\Gamma} \right), \\ \left(\frac{Q_{p,\Gamma}^2}{A_{p,\Gamma}} + g A_f h_{s,\Gamma} \right) - \left(\frac{Q_{f,\Gamma}^2}{A_{l,\Gamma}} + \frac{1}{2} \Lambda_{d,\Gamma} A_{l,\Gamma}^2 \right) = w_\Gamma \left(Q_{p,\Gamma} - Q_{f,\Gamma} \right), \end{gathered} \tag{7}$$

where the subscripts p and f denote the pressurized trace and the liquid projection of the stratified trace at the front, and $h_{s,\Gamma}$ follows from the elastic closure (6). The pressurized side is liquid-full, so gas does not cross the front;

the moving-frame gas condition gives

$$u_{g,\Gamma} = w_\Gamma. \tag{8}$$

Because the water-hammer speed a is much larger than mixed-flow front speeds, one pressurized acoustic characteristic always reaches the front, supplying the compatibility relation

$$u_{p,\Gamma} - u_{p,L} + \frac{g}{a}\,(h_{s,\Gamma} - h_{s,L}) = 0, \tag{9}$$

with L the adjacent pressurized state. Equations (7)–(9) do not determine all interface unknowns; the remaining information depends on the position of w_Γ relative to the stratified liquid signal speeds $s_\pm = u_{l,R} \pm c_{l,R}$, $c_{l,R} = \sqrt{\Lambda_{d,R} A_R}$, which partition the admissible front speeds into the slow ($w < s_-$), middle ($s_- \le w \le s_+$), and fast ($w > s_+$) active sets.

Because the correct active set is not known before w_Γ is computed, and branch-dependent nonlinear iteration at every front is fragile in transient computations, the closure of [12] is a fixed-cost one-step construction. Substituting the compatibility relation (9) and the mass jump into the momentum jump reduces the closure, for any candidate speed w, to a scalar quadratic for $A_{p,\Gamma}(w)$ and a scalar momentum residual $R(w)$. A finite-area predictor built from the adjacent states,

$$w_0 = \frac{A_{p,L} u_{p,L} - A_R u_{l,R}}{A_{p,L} - A_R}, \tag{10}$$

is projected onto each (slightly shrunken) active-set interval, one closed-form Newton-type residual correction is applied per interval, and the accepted candidate is the one minimizing a penalized criterion combining the Rankine–Hugoniot residual, the distance to its active set, and positivity penalties. No

Bernoulli-type energy relation is imposed at the front. The selected trace defines the moving-frame flux pair $\mathbf{G}_{p,\Gamma} = \mathbf{F}_{p,\Gamma} - w_\Gamma \mathbf{U}_{p,\Gamma}$ and $\mathbf{G}_{f,\Gamma} = \mathbf{F}_{f,\Gamma} - w_\Gamma \mathbf{U}_{f,\Gamma}$ passed to the cut-cell update below.

2.3. Finite-volume update and cut-cell front tracking

Away from fronts, both branches are advanced by the same explicit finite-volume machinery: MUSCL–TVD primitive reconstruction with the minmod limiter, branch-dependent Rusanov fluxes (scalar dissipation $\max(|u_p|+a)$ on the pressurized branch; block-diagonal dissipation with gas speed $|u_g|+c_g$ and stratified liquid speed $|u_l|+c_\Lambda$, $c_\Lambda = \sqrt{\max(\Lambda_d A_l, 0) + c_{\text{num}}^2}$, on the stratified branch), unsplit cell-centered sources, and SSP–RK2 time integration. The time step is the minimum of the branch CFL limits and a front-motion restriction $\Delta t_\Gamma = \frac{1}{2}\min(\Delta x_p, \Delta x_f)/|w_\Gamma^n|$, which prevents the front from sweeping more than half a cell per step.

The cell hosting a front is divided into pressurized and stratified sub-cells of lengths ℓ_p and ℓ_f with $\ell_p + \ell_f = \Delta x$. At the beginning of each step the adjacent branch states are passed to the closure of Section 2.2; the front is then advanced as $x_\Gamma^{n+1} = x_\Gamma^n + w_\Gamma \Delta t$, and each sub-cell is updated by its own branch flux at the outer face and the moving-frame interface flux at the front. Because both sub-cell balances use the same w_Γ and the same interface trace, the interface fluxes and swept-volume contributions cancel exactly for the conserved liquid components and gas mass: the front neither creates nor destroys liquid or gas. When the front crosses a cell face, the affected cells are remapped conservatively (event-based remap), and a bounded, localized residual relaxation corrects the remaining interface mismatch after an accepted step. Trapped gas regions are identified after each step as connected gas volumes bounded by liquid seals, and their pressures are evaluated from the isothermal equation of state on the current pocket mass and volume.

2.4. Vertical-shaft branch

A vertical shaft (ventilation tower, riser, or dropshaft barrel) is discretized as a separate one-dimensional finite-volume branch using the same area–momentum variables as the horizontal conduit, with the pipe inclination set to $\theta = 90°$ so that gravity acts axially. The shaft top is open to the atmosphere; the free surface inside the shaft and any gas core penetrating below it are represented by the axial profile of the gas volume fraction $\alpha_g(z)$, not by a prescribed bubble shape. The shaft state therefore distinguishes the free-surface elevation Y_{fs} (top of the continuous liquid column) from the air–water interface elevation Y_{int} (nose of the gas core rising through the column), which are exactly the two trajectories reported in shaft experiments. Geysering is diagnosed, following the experimental definition of [2], when Y_{fs} reaches the shaft top before Y_{int} catches the free surface.

2.5. Junction and local-loss closures

The shaft base is coupled to the horizontal conduit through a junction closure driven by the local pipe state and the trapped-pocket pressure. The junction exchange uses the pocket equation-of-state static head as the driving head, with the following local-loss elements retained from the frozen configuration used for all runs in Section 3:

- a junction entry/exit loss coefficient $K_j = 0.75$ on the through-flow between pipe and shaft base;

- an additional churn loss $K_g = 8.0$ applied while the junction carries gas, representing the strongly dissipative counter-current air-up/water-down exchange ("glugging") at the shaft mouth;

- a butterfly-valve loss $K_v = 2.0$ at the release valve, whose disc remains in the flow after opening, with a finite hand-opening time $t_v = 0.25$ s represented by throttling the loss coefficient during opening;

- a counter-current flooding (CCFL) limit of Wallis type on reverse liquid drainage at the shaft top while gas is venting, with limiting velocity scale $0.5\sqrt{gD_t}$, where D_t is the shaft diameter.

Three further elements govern the shaft base in the driven regime, i.e. when the trapped pocket holds an overpressure margin over the shaft column ($H_{a0} > Y_{fs,0}$) and gas has reached the junction; they are stated here and motivated with the narrow-tower test in Section 3.1.2: (i) the shaft base is driven by the pipe *crown* pressure (the elevation at which the shaft physically taps the pipe), one bore below the invert piezometric line while the junction is water-sealed; (ii) the shaft gas core connected to the pocket is pocket-fed—mouth influx raises the core nose, by the same fill-to-body-level closure used for the crown cavity in the horizontal pipe—rather than accumulating at the base cell under dispersed-bubble drag; and (iii) the entry gas flux is bounded by the Wallis counter-current flooding scale $j_g \leq C_w^2\sqrt{gD_t\,\Delta\rho/\rho_l}$ with $C_w = 0.9$. These closures use geometric facts and literature-scale constants; in the undriven regime (Test A of Campaign 1) they are inactive and the base retains the invert-datum coupling, a dual treatment whose consolidation is discussed in Section 4.

Once a trapped pocket becomes connected to the shaft and gas breaks through the liquid seal, venting latches on: the pocket continues to discharge through the shaft and is not re-sealed by intermittent slugs. This latch reflects the experimental observation that after breakthrough the release proceeds as a sustained, if oscillatory, discharge rather than a sequence of independent seal events.

2.6. Numerical configuration

All computations in Section 3 use one frozen solver configuration: horizontal grid spacing $\Delta s = 0.02$ m, shaft grid spacing $\Delta z = 0.01$ m, CFL number 0.35, and a reduced numerical elastic wave speed $a = 28$ m/s in the liquid-full branch. The reduced wave speed keeps the liquid-full reaches effectively incompressible on the pocket and shaft timescales (the gravity-wave scale is $\sqrt{gD} \approx 1$ m/s) while avoiding stiff acoustic ringing at grid-scale micro-voids; the geysering dynamics studied here are pocket-compression and gravity-wave driven, not water-hammer driven. Gas thermodynamics is isothermal with $R = 287$ J/(kg K) and $T = 293$ K, and the liquid density is $\rho_l = 998$ kg/m^3. No loss coefficient or numerical parameter is adjusted between test cases; the only case-dependent element is the regime-conditioned shaft-base treatment of Section 2.5 (crown-datum, pocket-fed, flooding-limited in the driven regime; invert-datum otherwise), which switches on the physical state, not on the test identity.

3. Test cases

The evaluation is organized as a sequence of test campaigns against published geysering experiments. Within each campaign, individual configurations are selected to span the experimentally observed regimes, and all configurations are computed with the single frozen solver configuration of Section 2.6. Simulations start from the documented experimental initial conditions; the shaft motion, the air release, and the regime selection are outcomes of the computation. No prescribed release rate, scripted geyser height, overflow accumulator, or per-case parameter fitting is used.

Three campaigns are selected from the experimental geysering literature so that each contributes a different kind of evidence. Campaign 1 (Section 3.1) reproduces the water-filled ventilation-shaft experiments of Vasconcelos and Wright [2]—the most widely cited controlled geysering experiments and the standard benchmark for geysering models—and tests the time-resolved histories in both regimes: Test A in the non-geysering branch and Test B in the geysering branch. Campaign 2 (Section 3.2) reproduces the horizontal-pipe/vertical-riser experiments of Cong, Chan and Lee [3], the only campaign that condenses its outcome into an explicit two-parameter geysering criterion; it therefore tests the model as a regime classifier across a systematic parameter sweep rather than in individual histories. Campaign 3 (Section 3.3) addresses the prototype-motivated junction-chamber experiments of Liu, Shao and Zhu [5], which connect the laboratory benchmarks to a field-scale dropshaft geometry. Other experimental studies—the field observations of [1], the larger-scale releases of [4] designed for three-dimensional CFD calibration, the covered-manhole geysers of [6], the mitigation study of [7], and the vertical-pipe violent geysers of [9]—are used as corroborating references rather than as reproduction targets, either because their geometry is not controlled at the level needed for one-dimensional comparison or because their focus (cover dynamics, mitigation devices, flashing-driven eruption) lies outside the entrapped-air release mechanism studied here.

3.1. Campaign 1: water-filled ventilation shafts (Vasconcelos and Wright, 2011)

The apparatus of [2] consists of a horizontal acrylic pipe of diameter $D = 0.094$ m with both far ends closed, a ventilation tower of length $L = 0.610$ m and variable diameter D_t mounted on a coupling 3.516 m from the upstream end, and an upstream air chamber of length 0.546 m separated from the water-filled reach by a butterfly valve (Fig. 1). The air chamber is pressurized to an initial gauge head H_{a0}, the tower is partially filled to an initial level $Y_{fs,0}$, and the run starts when the valve is opened by hand in less than one second. The pressurized pocket then propagates along the pipe crown toward the tower and vents through the water column inside it. A pressure transducer is located 1.07 m downstream of the valve. The experimental programme varied $D_t \in \{57.1, 44.4, 25.4, 12.7\}$ mm ($D_t/D = 0.607$–0.135), $H_{a0} \in \{0.305, 0.610, 0.915\}$ m, and $Y_{fs,0} \in \{0.254, 0.356, 0.457\}$ m. The controlling variable was found to be the tower diameter: for $D_t/D = 0.607$ the free-surface rise never exceeded 10% of L, whereas for $D_t/D = 0.135$ the free surface reached the tower top in every repetition.

Two configurations are selected to span the two regimes (Table 1). Following [2], results are non-dimensionalized by the tower length and the tower gravity-velocity scale: elevations and pressure heads as $Y^* = Y/L$ and $H^* = H/L$, and time as $T^* = t\sqrt{gD_t}/L$. Geysering is defined as the free surface Y^*_{fs} reaching the tower top ($Y^* = 1$) before the rising air–water interface Y^*_{int} catches the free surface.

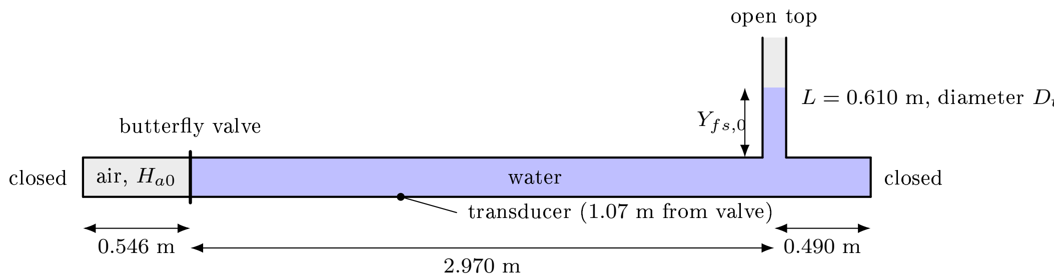

Figure 1: Schematic of the Vasconcelos and Wright [2] apparatus as discretized in the model (not to scale in the transverse direction). The horizontal pipe ($D = 0.094$ m) is closed at both far ends; the upstream chamber holds the pressurized air pocket; the ventilation tower (length $L = 0.610$ m, diameter D_t) is open at the top and partially filled to $Y_{fs,0}$.

Table 1: Selected test configurations from the experimental matrix of [2].

Test	D_t (mm)	D_t/D	H_{a0} (m)	$Y_{fs,0}$ (m)	$L/\sqrt{gD_t}$ (s)	Observed regime
A	57.1	0.607	0.305	0.356	0.815	no geysering
B	12.7	0.135	0.610	0.356	1.728	geysering (every run)

The model geometry follows the apparatus exactly: upstream air chamber 0.546 m (the initial pocket volume, 3.79×10^{-3} m^3), middle pipe 2.970 m, downstream closed pipe 0.490 m, tower at $x = 3.516$ m, and the transducer sampled at $x = 1.616$ m. The initial pocket pressure is $P_{a0} = P_{\text{atm}} + \rho_l g H_{a0}$; the pipe downstream of the valve and the tower below $Y_{fs,0}$ are initially at rest and water-filled. The experimental curves used for comparison were digitized from the published figures of [2] (pressure head from Figs. 5 and 6; free-surface and interface trajectories from Figs. 7 and 8, three experimental repetitions each); the digitized repetitions are shown either individually or as a band.

3.1.1. Test A: large tower, non-geysering branch ($D_t/D = 0.607$)

In the experiment this configuration vents the pocket without geysering: the tower free surface rises by less than 10% of L, the transducer head holds a long plateau near $H^* \approx 0.54$ while the pocket discharges, and the head then decays to zero once the pocket is exhausted. The computed run reproduces this branch. The free surface peaks at $Y^*_{fs} = 0.66$ (measured maximum 0.63), the rising gas core catches the free surface inside the tower, and no overflow occurs.

Figure 2 compares the transducer pressure head. The model reproduces the plateau level (cycle-averaged $H^* = 0.51$ against the digitized experimental plateau $H^* = 0.54$) over the full venting window $1 \lesssim T^* \lesssim 7.5$, and reproduces the terminal decay associated with pocket exhaustion, beginning at $T^* \approx 8.9$ in the model against 8.3 in the experiment. Two differences are visible. First, in $0 < T^* \lesssim 5$ the raw model trace carries a water-column–pocket spring oscillation with a period of about 0.7 s that decays by $t \approx 4$ s;

mass-conservation and phase checks identify it as a genuine volume–pressure oscillation of the release piston rather than a numerical artefact, but the experimental traces are more strongly damped. The plotted model curve is therefore cycle-averaged over a 0.8 s window. Second, the venting stage runs late and more gradually in the model: the measured head starts falling at $T^* \approx 8.3$ and reaches zero near 9.3, while the model holds its plateau to $T^* \approx 8.9$ and decays to about $H^* \approx 0.21$ by the end of the window. Both features reflect the slower discharge of the pocket through the tower once the (dissipative) cavity celerity feeds the vent at the reduced rate; the arrival-stage timing gains of the celerity value are traded against the venting-stage lag (Section 4).

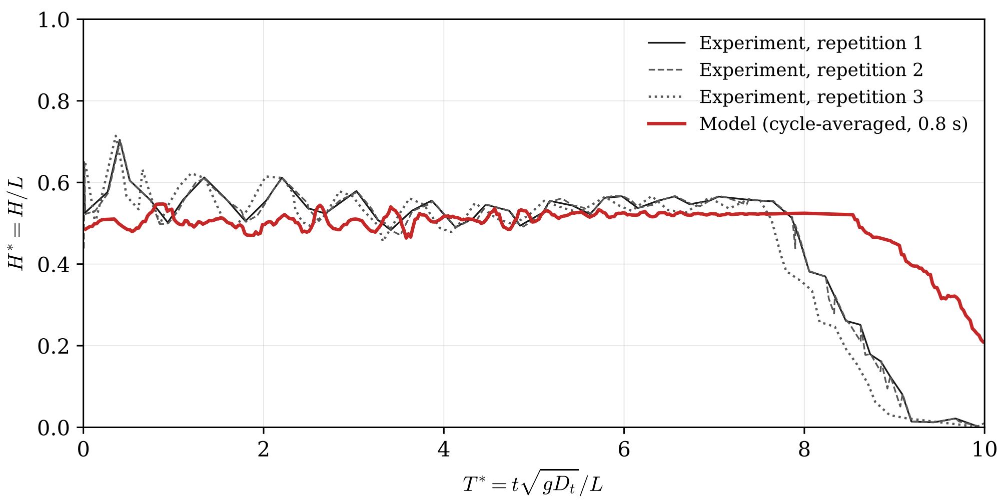

Figure 2: Test A ($D_t/D = 0.607$, $H_{a0} = 0.305$ m, $Y_{fs,0} = 0.356$ m): transducer pressure head H^* against T^*. Black lines: the three experimental repetitions of [2], digitized point by point from their Fig. 5; red line: model, cycle-averaged over a 0.8 s window.

Figure 3 compares the tower free-surface and air–water-interface trajectories in the published observation window ($T^* = 7$–10, Fig. 7 of [2]). The model holds the free surface near $Y^*_{fs} \approx 0.6$ throughout the venting plateau, consistent with the measured level, and produces the interface climb through the tower with the correct slope: the fitted non-dimensional interface velocity is $V^*_{int} = 0.41$, against the experimental Table-2 average of 0.39 and a photograph-based estimate of 0.50 for this tower size. The interface leaves the tower base at $T^* = 7.5$, inside the spread of the three experimental repetitions (climb onsets at 7.3, 7.8, and 7.9), and catches the free surface at $T^* = 8.8$ against the measured 8.4 (Table 2). Two systematic differences in the level histories are noted. First, the measured free surface creeps upward during the interface climb (from 0.59 to 0.65), which [2] attribute to the expansion of the depressurizing pocket as it occupies the tower; the computed level stays at its plateau value because the volume-neutral counter-current exchange at the mouth transits gas through the standing column without net volume addition, so the computed climb-phase free-surface velocity is $V^*_{fs} \approx 0$ against the measured 0.048 (the measured maximum level itself is matched: $Y^*_{fs,\max} = 0.66$ computed against 0.63 measured). Second, after the interface reaches the free surface, the measured tower level stays near $0.6L$ to the end of the observation window, while the model level decays faster once venting is established; the counter-current flooding limit at the tower mouth (Section 2.5) bounds but does not eliminate this too-fast drainage. Both differences trace to the same film-flow pocket topology discussed in Section 4.

Figure 4 shows computed tower snapshots at 0.14 s intervals during the interface climb, for comparison with the photographic sequence in Fig. 4

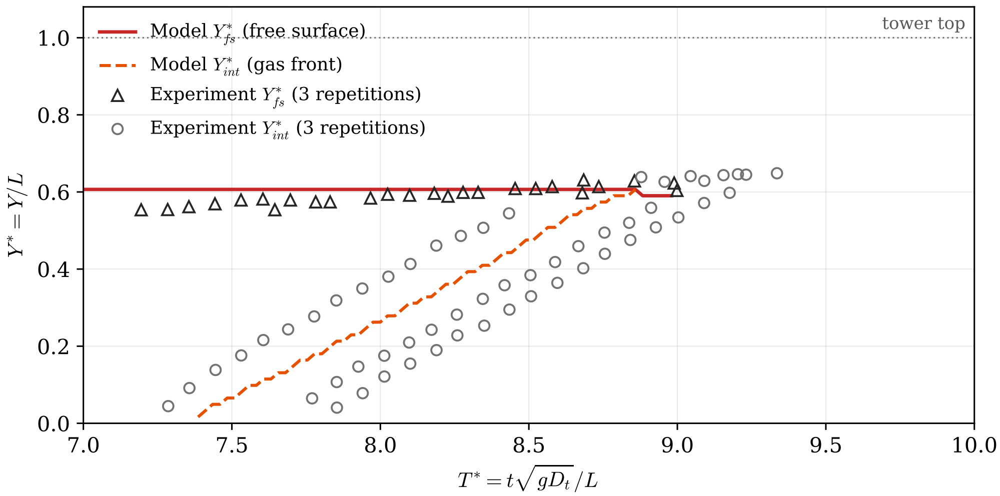

Figure 3: Test A: tower free-surface elevation Y^*_{fs} (solid) and air–water interface elevation Y^*_{int} (dashed) against T^* in the published observation window ($T^* = 7$–10). Symbols: the three experimental repetitions of [2], digitized from their Fig. 7 (triangles: free surface; circles: interface). The model gas-front curve ends where it catches the free surface (beyond that instant the two interfaces merge and a distinct front no longer exists); the model free-surface curve is shown over the extent of the experimental coverage.

of [2]. The gas core width is drawn from the local two-fluid gas fraction without thresholding. The computed front positions track the photographed front to within 0.03 m over the sequence, and the mean front speed over the photographed interval is 0.375 m/s in the experiment-based estimate against 0.31 m/s in the model fit window.

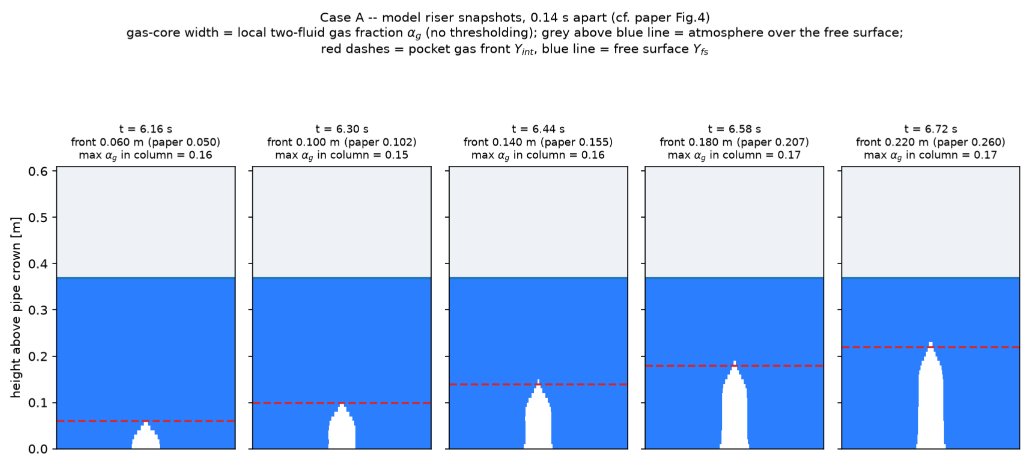

Figure 4: Test A: computed tower snapshots at 0.14 s intervals during the air–water interface climb (cf. the photographic sequence in Fig. 4 of [2]). Blue: water; white: gas core (local two-fluid gas fraction); grey: atmosphere above the free surface. The red dashed line marks the computed pocket gas front Y_{int} and the panel captions give the model and photographed front elevations.

Finally, Fig. 5 compares the present model directly against the numerical model of [2] themselves (their two-phase-analysis (TPA) model, a lumped pocket formulation with a Batchelor-type laminar film-flow closure). That comparison is reported for the neighbouring configuration $D_t/D = 0.607$, $H_{a0} = 0.305$ m, $Y_{fs,0} = 0.254$ m—the same tower and pocket pressure as Test A with a lower initial level—so the frozen solver is re-run at that initial

Table 2: Test A: measured against computed quantities. Times are non-dimensional ($T^* = t\sqrt{gD_t}/L$); velocities are normalized by $\sqrt{gD_t}$. Experimental values are digitized from [2] (Figs. 4, 5, 7 and Table 2).

Quantity	Experiment	Model
Regime	no geysering	no geysering
Maximum free-surface rise $Y^*_{fs,\max}$	0.63	0.66
Pressure-head plateau H^*	0.54	0.51
Onset of terminal head decay T^*	8.3	8.9
Interface liftoff from tower base T^*	7.3/7.8/7.9 (3 reps)	7.5
Interface catches free surface T^*	8.4	8.8
Interface climb velocity V^*_{int}	0.39 (Table 2), 0.50 (Fig. 4)	0.41
Climb-phase free-surface velocity V^*_{fs}	0.048	≈ 0

condition. The TPA-model curves and the experimental traces were digitized from Fig. 10 of [2] and are replotted here together with the present computation for the two panels that carry the discussion: the interface trajectory and the transducer head. Because the valve is opened by hand, a rigid time shift between the model and the published time axis would be legitimate; none is needed for the interface panel (the computed climb onset sits +0.2 from the published model curve, inside the repetition scatter) and none is applied. The interface trajectory Y^*_{int} reproduces the climb slope and span of both the measurements and the TPA model; the interface-velocity plateau ($V^*_{int} \approx 0.37$ computed, ≈ 0.4 TPA and measured, from the velocity panels of their Fig. 10) agrees; the pressure-head decay runs about 0.7 time units late (the venting-stage lag of the reduced cavity feed, as in Test A); and the head panel shows the structural difference already noted: the measured head and the TPA model decay *during* the climb, following the shrinking water column above the pocket nose, while the present model holds its plateau until breakthrough and then collapses. The TPA model obtains the during-climb decay by construction—its pocket is a lumped volume whose pressure is tied to the column above the nose through the film-flow closure—whereas in the present field formulation the tower base carries the weight of the full standing column as long as the gas transits it as a dispersed phase. Section 4 discusses this topological limitation and the closure experiment performed on it.

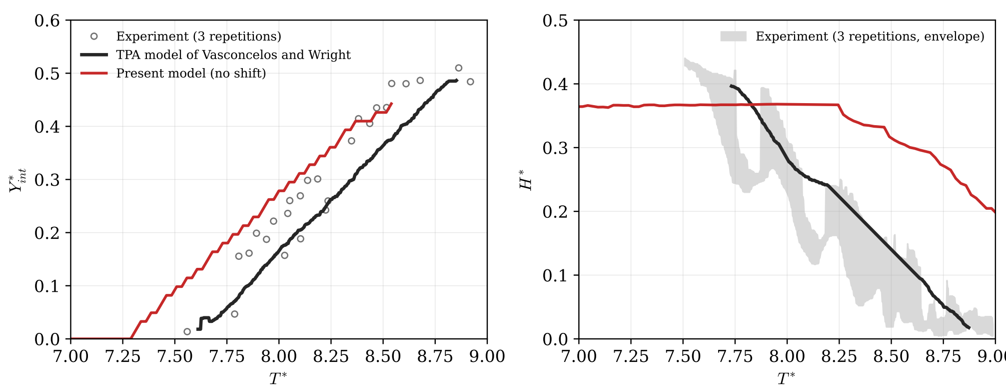

Figure 5: Campaign 1, configuration $D_t/D = 0.607$, $H_{a0} = 0.305$ m, $Y_{fs,0} = 0.254$ m: present model (red) against the TPA model of [2] (black) and their measurements (grey), both digitized from Fig. 10 of [2]. Left: air–water interface elevation Y^*_{int} (circles: experimental repetitions). Right: transducer pressure head H^* (shaded: envelope of the three experimental repetitions). No time shift is applied to the present-model histories (see text).

3.1.2. Test B: small tower, geysering branch ($D_t/D = 0.135$)

In the experiment this configuration geysered in every repetition: the tower level first rises to a quasi-static stand near $Y^*_{fs} \approx 0.82$ while the pocket propagates along the pipe crown, then the air–water interface enters the tower at $T^* \approx 3.65$ and the aerated free surface accelerates to the tower top at $T^* \approx 3.9$, before the interface (at the top near $T^* \approx 4.1$) catches it. The transducer head holds a plateau near $H^* \approx 0.76$ and collapses sharply at $T^* \approx 4.05$ when the pocket vents.

Three elements of the junction closure are specific to this narrow-shaft run and are stated before the comparison; each uses a geometric fact or a literature-scale constant, not a fitted value. (i) *Crown-datum base coupling*: the tower taps the pipe at its crown, so the shaft-base pressure is the crown pressure—one bore below the invert piezometric line while the junction is water-sealed—rather than the invert value; with the invert coupling the narrow tower is over-driven by up to $\rho_l g D$ ($0.15L$) and its column is pinned at the rim from the release transient onward. (ii) *Pocket-fed core*: while the pocket holds overpressure and the junction carries gas, the gas body in the shaft is connected to the pocket, so mass entering the mouth raises the core nose (the same fill-to-body-level closure used for the crown cavity in the horizontal pipe) instead of accumulating at the base cell, where the dispersed-bubble drag would cap the climb at the Taylor scale. (iii) *Flooding-limited entry*: the driven blow-through of a narrow shaft is bounded by the counter-current flooding (Wallis) scale, with superficial gas velocity $j_g \le C_w^2\sqrt{gD_t\,\Delta\rho/\rho_l}$ and $C_w = 0.9$ for a smooth tube end. The measured climb velocity $V^*_{int} = 1.43$ (Table 2 of [2]) is then the flooding-limited nose speed of the pocket-fed core;

no coefficient is tuned to it. Elements (ii) and (iii) are inactive in Test A, where the pocket holds no overpressure margin ($H_{a0} < Y_{fs,0}$); Test A uses the invert base coupling, and the consolidation of the two base-datum treatments into a single junction closure is discussed in Section 4.

With these closures the computed run reproduces the geysering branch quantitatively up to a uniform time offset (Figs. 6 and 7, Table 3). The free surface rises to the quasi-static stand $Y_{fs}^* = 0.84$ (measured 0.82)—the level at which the isothermal pocket expansion balances the tower column—and holds it until the interface enters the tower; the eruption then proceeds to the rim, as in every experimental repetition. The transducer plateau, read at the crown datum (the tap and tower-base elevation), is $H^* = 0.76$ against the measured 0.76; the invert-datum record (0.91) is retained in Fig. 6 as the faint trace, and the measured plateau is consistent with the crown-referenced reading of the isothermal pocket state combined with the measured tower level to within $0.02L$. The final free-surface climb velocity is $V_{fs}^* = 0.39$ against 0.44; the fitted interface velocity is $V_{int}^* = 0.71$ against the Table-2 average of 1.43—underpredicted by half, though still twice the Taylor-bubble scale 0.35 that a dispersed-drag closure would give, because the pocket-fed core inherits the reduced feed rate of the dissipative cavity celerity. The sharp plateau–collapse of the head at pocket venting is reproduced with the measured shape.

The remaining discrepancy is a stretching of the gas-phase sequence rather than a shift. The arrival events land within the repetition scatter: interface entry at $T^* = 3.52$ against 3.65, and the free surface at the top at 3.70 against 3.90. The subsequent venting-stage events run late: the interface reaches $0.85L$ at 4.44 against 3.85, and the head collapses at 4.49 against 4.05. The stretching is the same venting-stage lag as in Test A: the in-tower climb of the pocket-fed core and the pocket blowdown are both fed by the crown current, so they inherit the reduced celerity that centres the arrival timing. Figure 7 also shows the model trajectories rigidly shifted by +0.24 time units (the mean arrival offset) as a slope reference. The early-time model trace also carries the underdamped release-piston oscillation noted in Test A (the experimental traces damp within about one time unit; the schematic of [2] explicitly annotates this release oscillation), and the model column briefly splashes to the rim during the release swing before settling at the quasi-static stand.

Taken together, Tests A and B show that the framework discriminates the two experimental regimes of [2] from first principles—venting without overflow in the wide tower, eruption to the rim in the narrow tower—and reproduces the pressure plateaus, the pre-eruption stand and eruption of the narrow-tower column, the terminal venting decay, and the interface kinematics in both branches. The remaining biases are a consistent early arrival of the crown cavity (a fraction of a time unit in Test A, 0.5–0.7 units in Test B) and the underdamped release oscillation; the narrow-shaft closure elements introduced for Test B and their consolidation with the wide-tower configuration are discussed in Section 4.

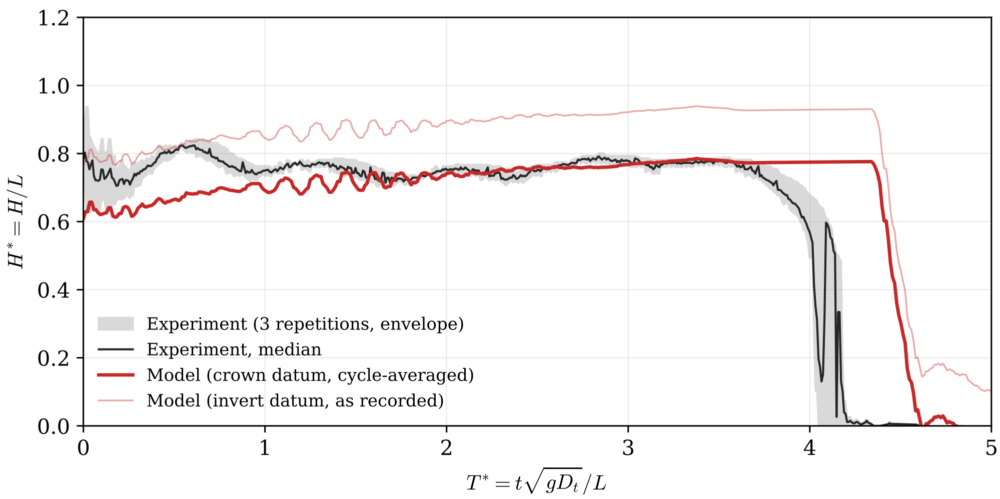

Figure 6: Test B ($D_t/D = 0.135$, $H_{a0} = 0.610$ m, $Y_{fs,0} = 0.356$ m): transducer pressure head H^* against T^*. Grey band and dark line: envelope and median of the three experimental repetitions, digitized from Fig. 6 of [2]; thick red line: model head cycle-averaged over a 0.4 s window and referenced at the crown datum (the transducer tap and tower-base elevation, $-D/L$ relative to the invert record); faint red line: the same trace at the invert datum as recorded.

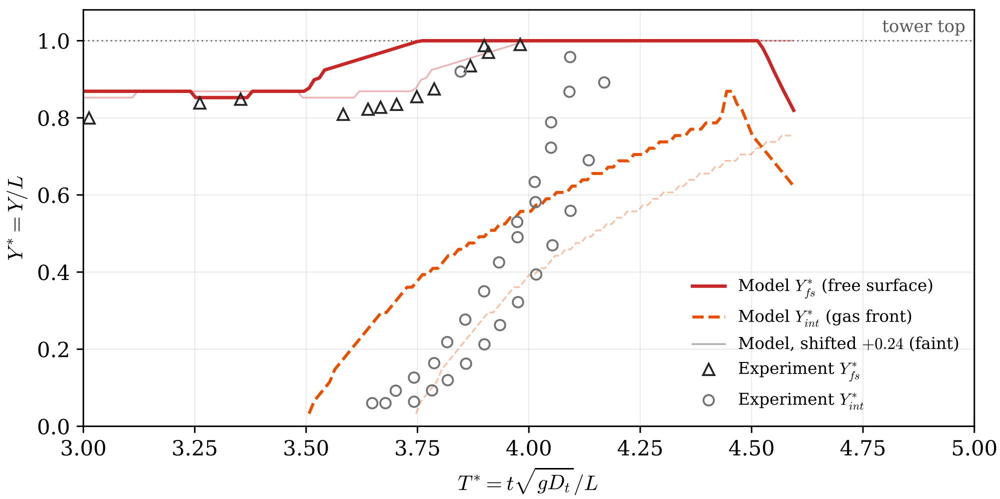

Figure 7: Test B: tower free-surface elevation Y^*_{fs} (solid) and air–water interface elevation Y^*_{int} (dashed) against T^* in the published observation window ($T^* = 3$–5). Symbols: digitized experimental points from Fig. 8 of [2] (triangles: free surface; circles: interface). The faint curves repeat the model trajectories rigidly shifted by $+0.24$ time units (the mean arrival offset) as a slope reference; model curves are shown over the extent of the experimental coverage.

Table 3: Test B: measured against computed quantities. Times are non-dimensional ($T^* = t\sqrt{gD_t}/L$); velocities are normalized by $\sqrt{gD_t}$. Experimental values are digitized from [2] (Figs. 6 and 8, and Table 2). The model pressure plateau is read at the crown datum (invert record 0.91).

Quantity	Experiment	Model
Regime	geysering (every run)	geysering
Pre-eruption stand of the free surface Y^*_{fs}	0.82	0.84
Pressure-head plateau H^* (crown datum)	0.76	0.75
Interface climb velocity V^*_{int}	1.43	0.71
Free-surface climb velocity V^*_{fs}	0.44	0.39
Interface enters tower T^*	3.65	3.52
Free surface reaches tower top T^*	3.90	3.70
Interface reaches $0.85L$ T^*	3.85	4.44
Sharp head drop at pocket venting T^*	4.05	4.49

3.2. Campaign 2: entrapped-air release from a horizontal pipe into a vertical riser (Cong, Chan and Lee, 2017)

3.2.1. Experimental apparatus and test conditions

The apparatus of [3] consists of a horizontal acrylic pipe of diameter $D = 0.05$ m and length approximately 6 m, connected at its upstream end to a constant-head tank and carrying a vertical riser of height 1.8 m and variable diameter ($D_r = 16$–46 mm) mounted through a tee-junction. An air pocket of prescribed initial length L_0 is created in the horizontal pipe between ball valves; releasing it under the upstream head H_0 drives the pocket along the pipe crown to the riser, where it vents through the initially water-filled column. Air–water interface trajectories in the pipe and riser were measured by video and high-speed camera, and pressures near the pipe end and at the riser base by transducers. From their parameter study the authors condensed the outcome into a two-parameter geysering criterion,

$$\frac{D_r}{D} \leq 0.62 \qquad \text{and} \qquad V^*_{air} = \frac{V_{air}}{(\pi D_r^2/4)\, H_0} \geq 3.42, \tag{11}$$

i.e. geysering requires both a sufficiently narrow riser and a pocket volume large relative to the riser water column.

This campaign therefore evaluates the model in a different role from Campaign 1: not against individual time histories, but as a regime classifier over a systematic sweep. Two comparisons are made. First, the seven high-speed-camera runs of the experimental Series B (B-H1 – B-H7: $H_0 = 0.66$ m, $L_0 = 0.61$ m, riser diameters $D_r/D = 0.32$–0.92) are computed individually and compared against the measured classification, pocket arrival times, and mean rise velocities. Second, a broader synthetic sweep over the experimental parameter ranges ($D_r = 16$–46 mm, $L_0 = 0.61$–1.8 m, $H_0 = 0.66$–0.88 m; 63 configurations) is computed and the model classification is compared against the experimental criterion, Eq. (11).

All runs of this campaign use the same fully synchronous junction coupling as Campaign 1, with the identical frozen solver configuration (the Campaign-1 Test-B closure set; no parameter is changed between campaigns or between runs). Two geometric elements are specific to the apparatus: the upstream boundary is the constant-head tank, and the pocket is created *downstream* of the riser tee, so the released pocket travels upstream along the crown toward the riser. The tee position (2.88 m from the tank) is fixed from the experiment's own kinematics—the measured arrival times and front speeds of Table 2 of [3] give a valve-to-tee distance $T_a U_f$ of 2.51/1.92/1.32 m for the three pocket lengths $L_0 = 0.61/1.2/1.8$ m, all placing the tee at the same station—and is held fixed across the sweep. An earlier version of this campaign used a staged horizontal/vertical treatment whose classifier reduced to a static volume–diameter criterion; those results were circular as a validation of Eq. (11) and are superseded by the fully coupled results reported here.

3.2.2. Series B: classification, arrival times, and rise velocities

Table 4 and Fig. 8 summarize the seven Series-B runs. The model reproduces the measured classification in five of the seven: geysering for B-H1 ($D_r/D = 0.32$) and no geysering for B-H4–B-H7 ($D_r/D = 0.62$–0.92), with no false positives. The two intermediate geysering runs B-H2 and B-H3 ($D_r/D = 0.42$, 0.52) are misclassified as non-geysering: the computed free surface is driven to 0.87 and 0.82 m but does not reach the rim. The miss is one-sided—the model errs only toward under-predicting eruption—and its mechanism is identified in Section 4: the volume-neutral mouth exchange, which is what lets the narrow-riser pocket rebuild the overpressure that powers the B-H1 geyser, also cancels part of the one-to-one column lift in the reservoir-fed layout, and the deficit grows with riser width. The computed arrival times are late by 1.5 s on average (largest for the widest riser, where the arrested crown current covers the final reach slowly), against a measured spread of 7.84–8.22 s.

In the geysering run B-H1 the computed eruption-window climb speeds (v_{fs} = 1.71, v_{int} = 1.28 m/s as the fastest sustained 0.6-s climb) bracket the measured whole-event means (0.92, 1.23 m/s): the eruption jet itself is over-sharp, the same narrow-riser bias identified in Test B of Campaign 1. In the no-geysering branch the computed gas-nose climbs (0.69–1.20 m/s) exceed the measured 0.42–0.48 m/s because the computed climb concentrates in surge pulses rather than the steady Taylor-scale ascent observed; the computed free-surface response (0.20–0.65 m/s during the pulses, zero sustained rise) shows the same pulsed character against the measured steady 0.2–0.26 m/s.

The computed pipe-end pressure transient now differentiates the branches under the synchronous coupling: the geysering case B-H1 develops a compression surge to $1.71\,H_0$ (cycle-averaged) against about $1.9\,H_0$ reported, while the no-geysering case B-H6 shows only a gentle post-arrival hump to $1.02\,H_0$ against about $1.4\,H_0$ reported. The staged treatment gave both runs the same $2.07\,H_0$ peak; recovering the experimental differentiation—a wide riser vents the pocket and thereby damps the pipe-end pressure—is the principal gain of the synchronous coupling, obtained at the cost of the two intermediate-diameter classification misses discussed above.

Table 4: Campaign 2, experimental Series B of [3] ($H_0 = 0.66$ m, $L_0 = 0.61$ m): measured against computed classification, pocket arrival time T_a, maximum computed free-surface height $Y_{fs,\max}$ (riser top at 1.8 m), and rise speeds of the free surface (v_{fs}) and gas nose (v_{int}). Measured speeds are whole-event means (Table 2 of [3]); computed speeds are the fastest sustained 0.6-s climb in the venting window. Misclassified runs are marked †.

			Geysering		T_a (s)			v_{fs} (m/s)		v_{int} (m/s)
Run	D_r/D	V^*_{air}	Meas.	Model	Meas.	Model	$Y_{fs,\max}$ (m)	Meas.	Model	Meas. / Model
B-H1	0.32	9.03	yes	yes	8.07	7.56	1.80	0.92	1.71	1.23 / 1.28
B-H2†	0.42	5.24	yes	no	7.84	–	0.87	0.77	–	1.02 / –
B-H3†	0.52	3.42	yes	no	8.18	9.56	0.82	0.66	0.43	0.92 / 1.24
B-H4	0.62	2.40	no	no	8.14	9.40	0.82	0.21	0.53	0.42 / 1.20
B-H5	0.72	1.78	no	no	8.10	9.88	0.86	0.26	0.65	0.48 / 1.05
B-H6	0.82	1.37	no	no	8.10	9.16	0.78	0.25	0.39	0.48 / 1.05
B-H7	0.92	1.09	no	no	8.22	11.44	0.79	0.20	0.20	0.44 / 0.69

Figure 9 shows the two signature runs in detail (these are the runs examined as individual time histories in Sections 3.2.2 and the case folders; the per-run comparisons against the digitized level and pressure histories of [3] Figs. 7, 9 and 10 are reported with the archived reproduction). In B-H1 ($D_r/D = 0.32$) the released pocket arrives at 7.56 s (measured 8.07 s), the entering gas drives the free surface up in two surge pulses, and the arrest of the supply-line counter-flow compresses the pocket to 1.71 H_0 (cycle-averaged; reported $\approx 1.9\,H_0$), ejecting the column—the free surface reaches the rim with the gas nose still below it, the geysering signature. In B-H6 ($D_r/D = 0.82$) the same release delivers the same pocket, but the wide riser passes the gas with a gentle push of the free surface to 0.78 m and the pocket vents without

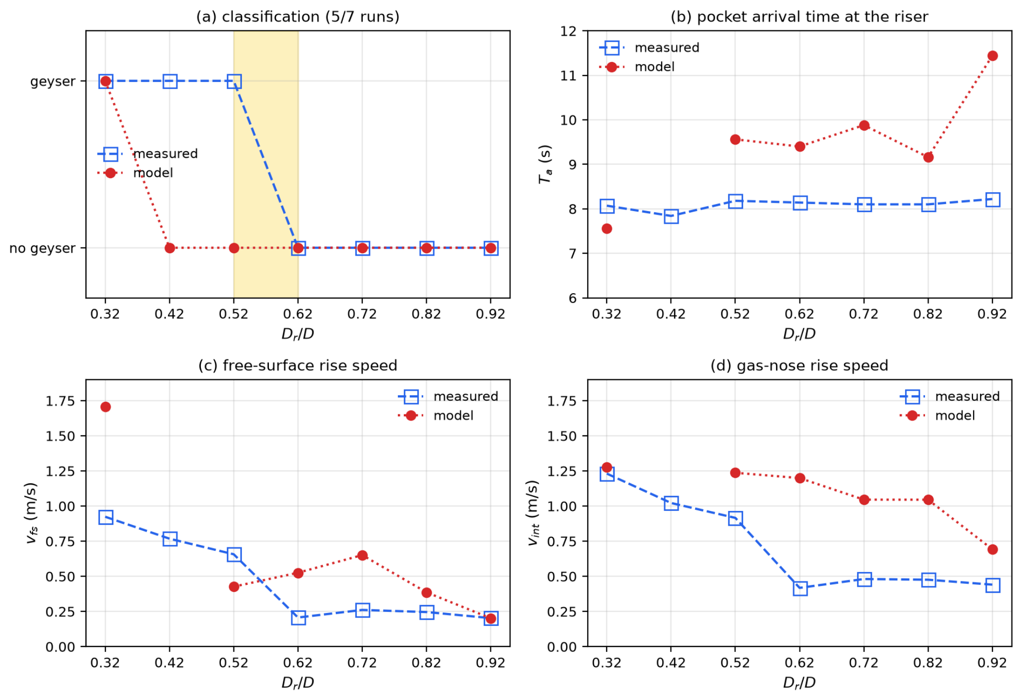

Figure 8: Campaign 2, Series B of [3]: measured against computed (a) geysering classification, (b) pocket arrival time at the riser, (c) mean free-surface rise speed, and (d) mean gas-nose rise speed, as functions of the riser-to-pipe diameter ratio D_r/D.

eruption; its pressure never exceeds $1.02\,H_0$ after arrival. The branch differentiation of the pipe-end pressure—the experiment's cleanest controlled contrast, since only D_r changes—is reproduced in sign and ranking, though both levels are under-predicted (the $1.4\,H_0$ no-geyser hump is missed because the computed free-surface lift is under-predicted in the reservoir-fed layout, Section 4).

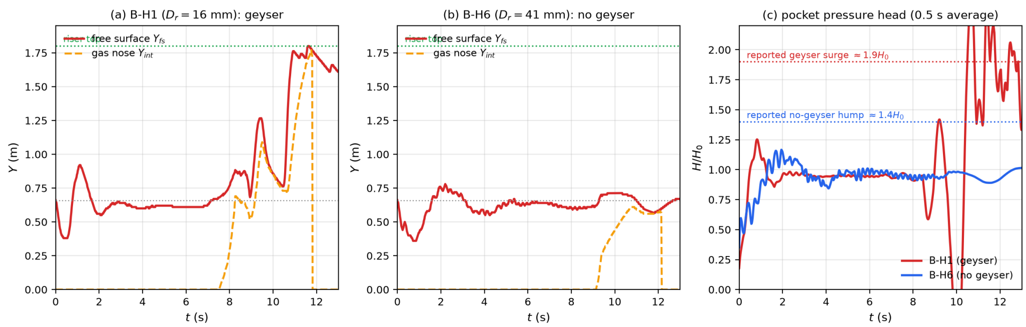

Figure 9: Campaign 2, signature runs of [3] under the fully synchronous coupling: (a) computed riser free-surface and gas-nose trajectories for the geysering run B-H1 ($D_r = 16$ mm); (b) the same for the no-geysering run B-H6 ($D_r = 41$ mm); (c) computed pocket pressure head (0.5 s average) for both runs against the reported peak levels ($\approx 1.9\,H_0$ geyser surge; $\approx 1.4\,H_0$ no-geyser hump).

3.2.3. Regime map against the experimental criterion

Figure 10 shows the 63-configuration sweep in the criterion plane (D_r/D, V^*_{air}). Each configuration is one full synchronous simulation classified with no knowledge of Eq. (11); the criterion lines are drawn for comparison only. The model classification agrees with the experimental criterion in 39 of 63 configurations, and the 24 disagreements are strictly one-sided: in every one the criterion predicts geysering and the model does not. There are no false positives—every configuration the model erupts lies inside the criterion's geysering region, and the no-geysering half-plane $D_r/D > 0.62$ is classified correctly throughout, as is the small-volume region $V^*_{air} < 3.42$. The misses concentrate at intermediate diameters ($D_r/D = 0.42$–0.62) combined with large pocket volumes ($L_0 \geq 1.2$ m), where the computed free surface is driven to 1.0–1.6 m but falls short of the 1.8-m rim; seven of the 24 reach within 0.45 m of it. The pattern is the Series-B under-lift bias extended over the parameter plane: the model captures the physical structure of the regime boundary—both end members and the interaction sense that eruption requires narrow risers *and* large relative pocket volume—but places the eruption threshold conservatively, under-predicting eruption in the transition band. For screening use the bias direction matters: the model as frozen errs toward missing geysers near the boundary, not toward false alarms.

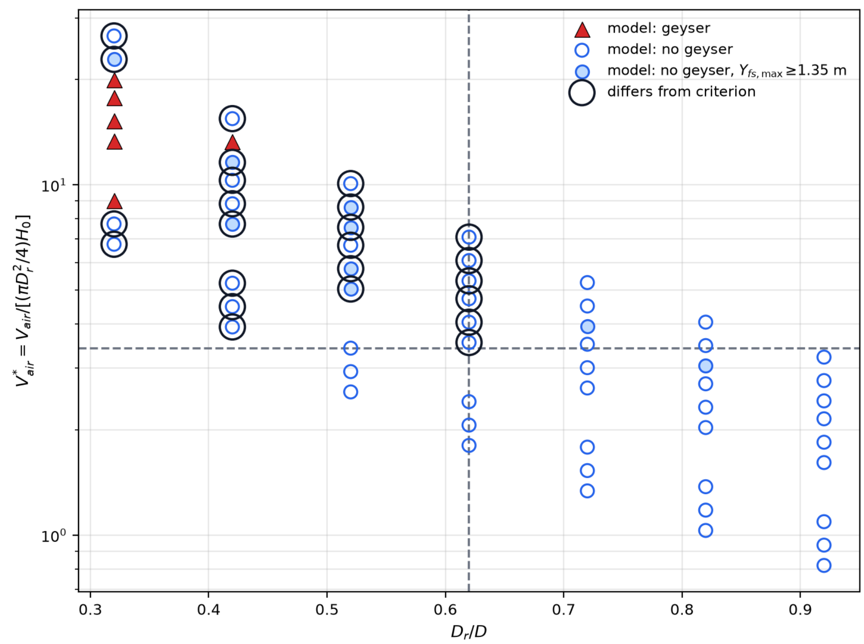

Figure 10: Campaign 2: model geysering classification for 63 configurations spanning the experimental parameter ranges of [3] (D_r = 16–46 mm, L_0 = 0.61–1.8 m, H_0 = 0.66–0.88 m), each a full synchronous simulation, plotted in the criterion plane of Eq. (11). Filled triangles: model geysers; open circles: model does not geyser (filled light circles: free surface within 0.45 m of the rim); black rings: disagreement with the criterion; dashed lines: the criterion thresholds $D_r/D = 0.62$ and $V^*_{air} = 3.42$. The model agrees in 39 of 63 configurations; all 24 disagreements are criterion-geyser/model-no, with no false positives.

3.3. Campaign 3: stormwater geyser in a vertical shaft above a junction chamber (Liu, Shao and Zhu, 2020)

3.3.1. Experimental apparatus and one-dimensional representation

The apparatus of [5] is a 1:20-scale model of a 27 m deep Edmonton prototype dropshaft. A 5.80 m upstream pipe ($D = 0.20$ m, slope 1:100) feeds a transparent junction chamber ($0.30 \times 0.30 \times 0.45$ m) with an invert drop of 0.18 m to a 5.95 m downstream pipe ($D_d = 0.28$ m). A vertical shaft of diameter $d_r = 0.06$ m and height 1.22 m is mounted at the chamber lid centre and vents to the atmosphere. Inflow is imposed as a flow step $Q_0 \to Q_1$ with valve opening time $T_v \approx 0.4$ s. Pressure transducers PT1–PT4 monitor the shaft, the chamber lid (PT2, the primary datum), the chamber floor, and the upstream crown.

Three configurations are computed with the same frozen solver as Campaigns 1–2. The horizontal reach is represented as a single composite one-dimensional domain—upstream pipe, chamber, and downstream pipe—with internal momentum faces at the chamber connections and a rigid-column ordinary differential equation for the shaft water column coupled at the lid tap. This is a reduced one-dimensional formulation: three-dimensional chamber mixing, slug venting at the chamber gas port, and the arrival of a compressing air pocket along the upstream crown in the late-time regime are not resolved at the level of the companion two-fluid/RTRP framework and are stated explicitly where they limit the comparison.

3.3.2. Branch pair A2/B3: downstream boundary flips the outcome

Cases A2 and B3 share the same inflow step ($Q_0 = 20$, $Q_1 = 100$ L/s) and differ only in the downstream tail: A2 discharges to an open channel (weir-controlled, $h_d/D_d = 1/4$), whereas B3 is initially surcharged with a submerged tail gate. In the experiment A2 never geysered while B3 produced a single-shoot eruption on every repetition (Fig. 5 of [5]).

The model reproduces this branch flip without retuning. For A2 the riser column remains below the rim ($h_{r,\max} = 0.47$ m against a shaft height of 1.22 m) and the steady chamber pressures agree with the digitized Fig. 3 record: PT3 5.26 kPa against 4.99 kPa (-5%) and PT2 1.80 kPa against 2.15 kPa ($+16\%$ in the standing-column head that emerges from the composite domain rather than being prescribed). For B3 the computed run geysered, with the free surface reaching the shaft top and overflow. The eruption timing is close—peak PT2 at 1.68 s against 1.47 s—but the compressive peak amplitudes are underpredicted with the frozen elastic branch celerity $a_{wh} = 40$ m/s: PT2 peak 31 kPa against 55 kPa (-44%). A sensitivity run at $a_{wh} = 80$ m/s gives 63.5 kPa, confirming that the deficit scales with the acoustic wave speed rather than with a missing branch-specific closure. The computed jet height $h = 2.17$ m lies on the experimental regression line of Fig. 7(a) ($h = 0.694\, P_{\max}/\rho g + 0.309$ m, $R^2 = 0.97$).

3.3.3. Case C9: transient-driven geysers and the analytic mass-oscillation model

Case C9 ($Q_0 = 25 \rightarrow Q_1 = 40$ L/s, surcharged downstream tail, initial shaft column $h_{r0} = 0.30$ m, sealed air wedge on the upstream crown in the experiment) is the authors' detailed two-stage case: the first two geysers are driven by the hydraulic transient before the trapped pocket reaches the chamber at $t \approx 6.46$ s; geysers 3–8 are pocket-driven and lie outside the present reduced solver.

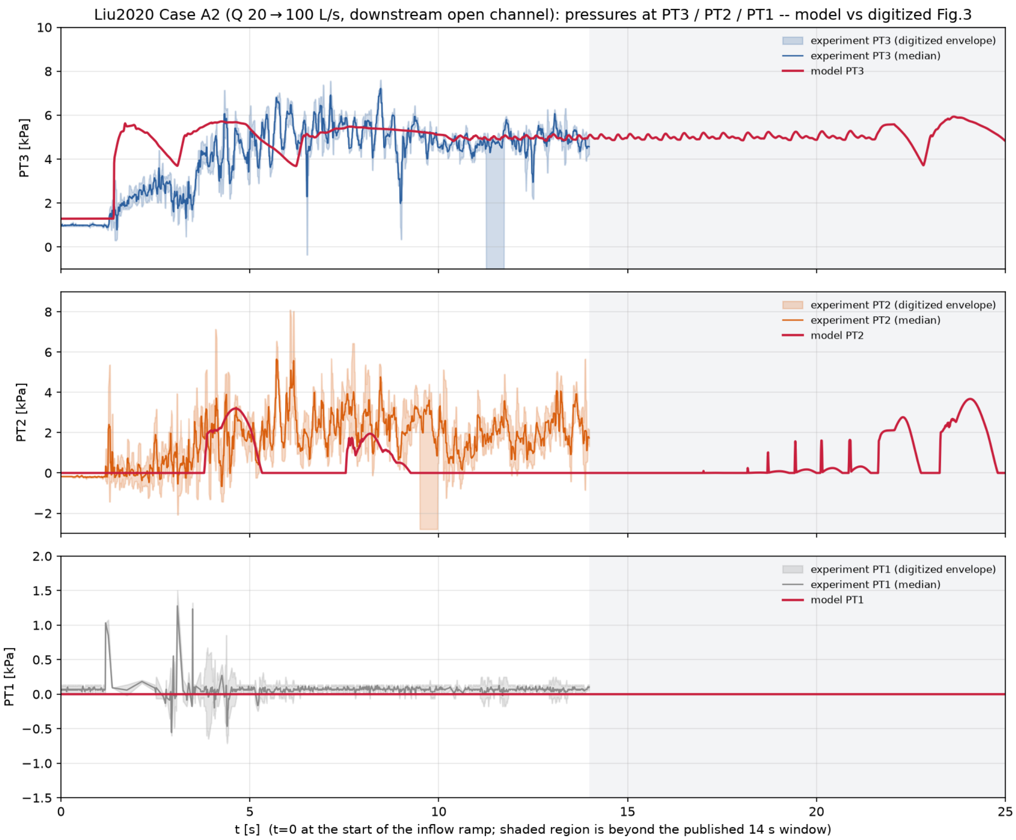

Figure 11: Case A2 (Q: 20→100 L/s, downstream open channel): chamber and shaft pressures against time. Blue: digitized experimental PT2/PT3 from Fig. 3 of [5]; red: present model.

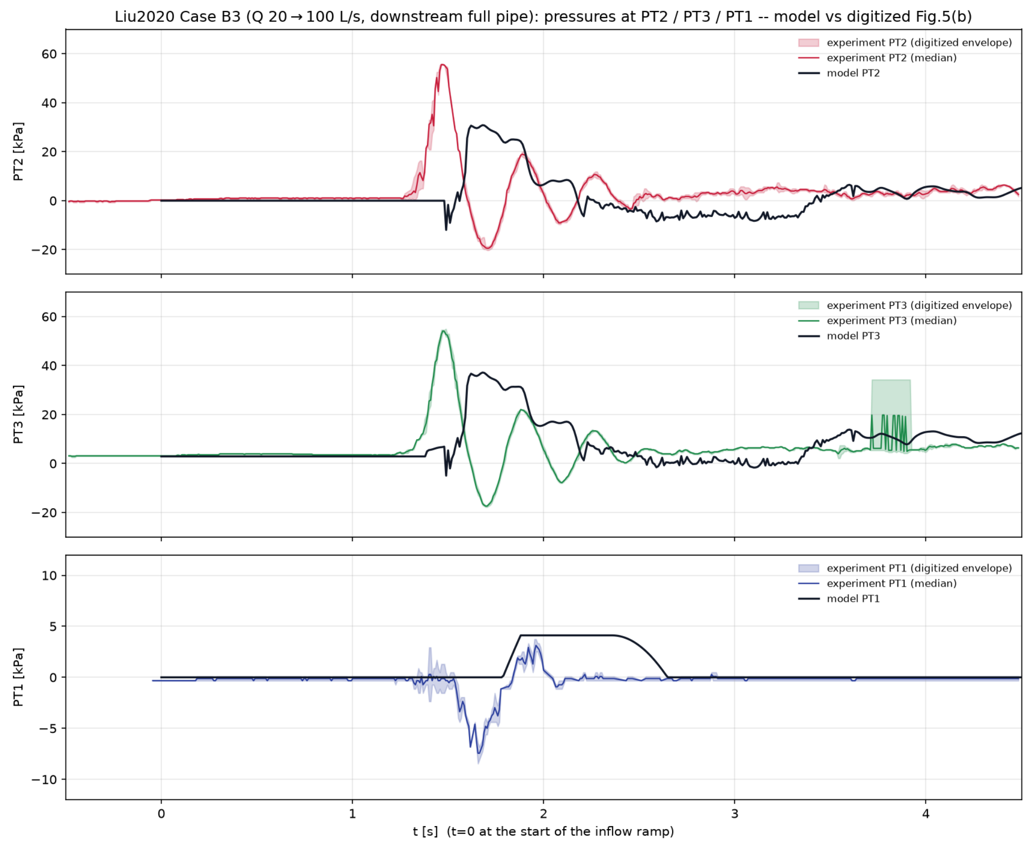

Figure 12: Case B3 (Q: 20→100 L/s, downstream surcharged): PT1–PT3 against time. Blue: digitized Fig. 5(b) of [5]; red: present model. The model selects the geysering branch; compressive peaks are acoustically under-resolved at the frozen $a_{wh} = 40$ m/s.

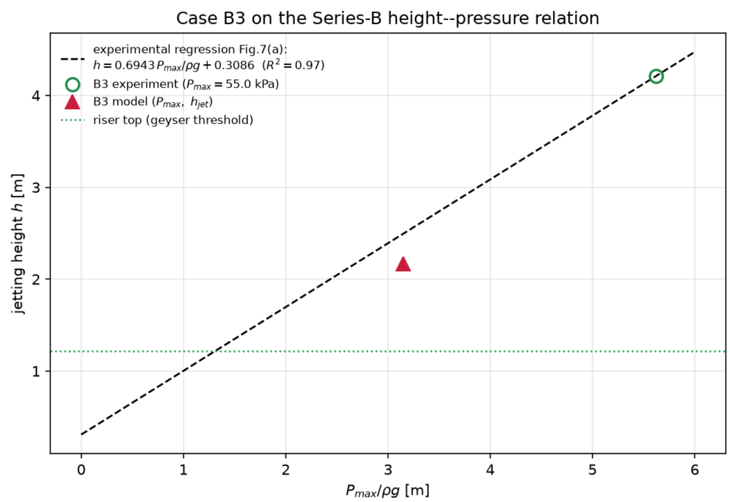

Figure 13: Case B3 on the eruption-height–peak-pressure diagram of Fig. 7(a) of [5]. Open symbols: experimental family; filled red circle: present model ($h = 2.17$ m, $P_{\max} = 31$ kPa at $a_{wh} = 40$ m/s).

For the phase-one comparison the pocket is omitted in the computation (the all-water variant assumed in Eq. (7) of [5]), while the sealed-pocket run is retained as a qualitative cushioning check. The analytic head at the chamber lid is

$$H_s(t) = h_{r0} + \frac{L_d}{gA_d}\frac{\mathrm{d}Q_u}{\mathrm{d}t}\left(1 - \cos\frac{2\pi t}{T}\right), \qquad T = 2\pi\sqrt{\frac{L_d A_r}{gA_d} + \frac{h_{r0}}{g}}, \tag{12}$$

with $T = 1.52$ s for C9.

Figure 14 compares the digitized PT2 record of Fig. 9 (converted to head at the chamber-top datum) against Eq. (12) and the all-water model over the first 5 s. The first peak is reproduced in timing and magnitude: measured $P_{1m} = 10.69$ kPa at $t = 0.50$ s, model 11.5 kPa at 0.54 s; the free surface reaches the shaft top at 0.73 s in the experiment and 0.93 s in the model. The oscillation period is $T = 1.45$ s in the paper (Eq. (8)) against 1.13 s in the model and 1.52 s in Eq. (12); the model is closer to the analytic value than to the measured period but still fast, consistent with the stiff one-dimensional tail-gate coupling. Steady heads after the first transient agree within $\approx 20\%$ (PT2 final 8.79 vs 10.6 kPa). The sealed-pocket variant delays and cushions the first peak as in the experiment qualitatively, then drifts after ≈ 4 s when the wedge compresses through the reduced gas model—declared out of scope for phase two.

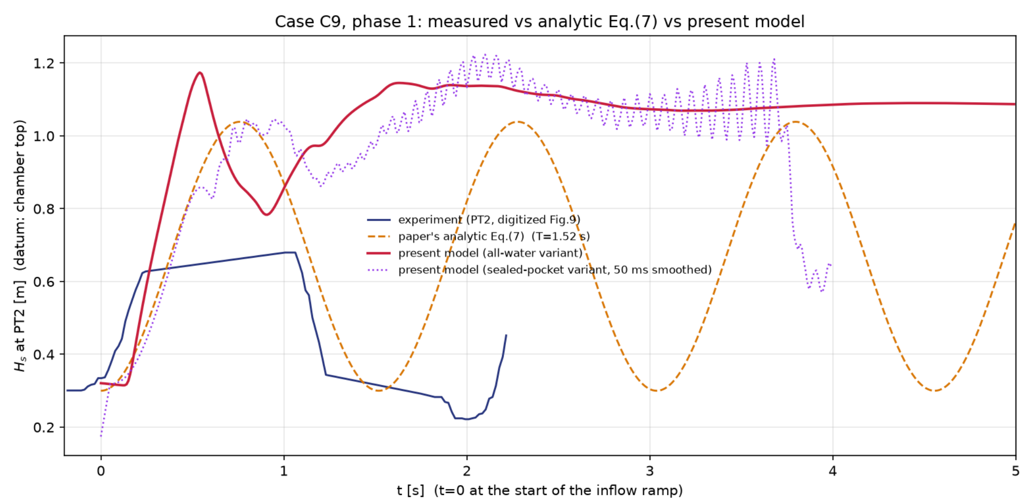

Figure 14: Case C9, phase 1 ($t < 5$ s): head H_s at PT2 (chamber-top datum). Blue: digitized Fig. 9 of [5]; orange dashed: Eq. (12); red: present model (all-water variant); purple dotted: sealed-pocket variant (50 ms smoothed). Pocket-driven geysers after $t \approx 6.5$ s are not computed.

Taken together, the three cases show that the composite-domain formulation extends the frozen Campaign 1–2 solver to a prototype-motivated junction geometry: a downstream-boundary branch pair (A2/B3) and a transient mass-oscillation geyser (C9 phase 1) are reproduced at the level of regime selection, timing, and—where the physics is hydraulic rather than

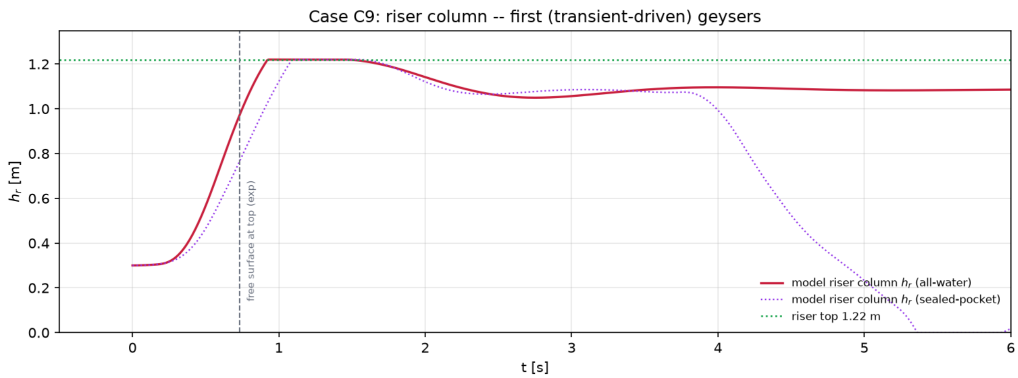

Figure 15: Case C9: shaft water-column height h_r for the all-water (solid) and sealed-pocket (dotted) variants. The dashed line marks the experimental time at which the free surface first reaches the shaft top in phase 1.

Table 5: Campaign 3 summary metrics. Acoustic peaks use the frozen $a_{wh} = 40$ m/s; no coefficient is retuned between cases.

Quantity	A2 (no geyser)	B3 (geyser)	C9 phase 1
Geyser?	no / no	yes / yes	yes / yes
PT2 steady or peak [kPa]	1.80 / 2.15	31 / 55	11.5 / 10.7
Peak or top-arrival time [s]	—	1.68 / 1.47	0.54 / 0.50
Shaft top reached [s]	—	≈ 1.65	0.93 / 0.73
Oscillation period [s]	0.30 (PT2–PT3)	—	1.13 / 1.45

Each cell: model / experiment.

acoustic—amplitude. The principal residual in this campaign is compressive under-resolution of short pressure spikes when the frozen elastic celerity is used, and the deliberate omission of pocket-transport physics after the pocket arrives at the chamber.

4. Discussion

4.1. What the framework resolves

The comparisons in Section 3 indicate that a one-dimensional formulation that conserves gas mass and momentum in the shaft and pipe, and that treats trapped pockets by an equation of state on the resolved pocket volume, is sufficient to select the geysering regime without regime-specific inputs. In the wide-tower test the venting pathway—pocket discharge through the tower with counter-current exchange at the mouth—sets a pressure plateau whose level and duration are controlled by the pocket volume and the tower hydrostatic column; both are reproduced because they follow from conservation rather than from calibrated closures. The interface climb velocity in the wide tower is likewise reproduced at the Taylor-bubble scale without any drift-flux input, because the buoyant rise of the gas core through the resolved liquid column is part of the two-fluid solution.

The regime discrimination between Tests A and B is controlled by the tower diameter through two competing rates that the model resolves: the rate at which the pocket discharges gas into the tower base, and the rate at which the tower can pass gas through its water column and vent it at the rim. In the wide tower the column passes the gas with a modest surface rise; in the narrow tower the same discharged volume displaces the column to the rim. This is the mechanism identified experimentally in [2], and it emerges here from the computed state.

Campaign 2 tests whether this discrimination generalizes beyond a single apparatus, and the answer is structured rather than uniform. Across the 63-configuration sweep of the [3] parameter ranges, computed with the same frozen solver and no knowledge of the experimental criterion, the model reproduces both end members of the regime boundary exactly—every wide-riser configuration ($D_r/D > 0.62$) and every small-volume configuration ($V^*_{air} < 3.42$) vents without eruption, and the narrowest risers with large pockets erupt—and it reproduces the interaction sense of the criterion, since both thresholds derive from the same resolved competition: the pocket must sustain its overpressure against the shrinking column (volume condition) faster than the riser can vent the gas through the column (diameter condition). Inside the transition band, however, the model is conservative: 24 of 63 configurations that the criterion marks as geysering are computed as strong but sub-rim column lifts (1.0–1.6 m against the 1.8-m rim), and the corresponding two intermediate Series-B runs (B-H2, B-H3) are the only classification misses among the seven high-speed-camera runs. All disagreements are one-sided—no configuration erupts in the model that the criterion excludes—so the frozen configuration errs toward missing marginal geysers, not toward false alarms.

4.2. Identified limitations

The completed campaigns expose five limitations, all at the level of the junction and closure treatment rather than of the two-fluid formulation, and all consistent with the model simplifications listed in the companion paper

[12].

Junction base datum. The two branches of Campaign 1 currently use different elevation datums for the shaft-base coupling: the wide-tower run transmits the invert piezometric head to the shaft base, while the narrow-tower run transmits the crown pressure (the tower physically taps the pipe crown, one bore above the invert). The crown datum is the geometrically correct choice, and in the narrow tower the difference, up to $\rho_l g D = 0.15L$ of spurious driving head, decides between a column pinned at the rim and the measured quasi-static stand. In the wide tower the distinction is masked by the aerated column (the visible level includes gas-holdup swell) and the invert coupling reproduces the measurements; a single junction closure that carries the vertical pressure distribution across the mouth—rather than one scalar datum—is the consolidation target.

Narrow-shaft climb closure. The measured interface climb in the narrow tower ($V^*_{int} = 1.43$, far above the Taylor-bubble scale) is approached only when the shaft gas connected to the pocket is treated as a pocket-fed core whose nose is raised by the mouth influx, with the influx bounded by the Wallis counter-current flooding scale; with the dissipative cavity celerity the computed nose speed is 0.71—half the measured value, but still twice the Taylor scale—because the mouth influx inherits the reduced crown feed. These two elements use literature constants ($C_w = 0.9$) and geometric closure levels, not per-case fits, but they are engaged only in the driven regime ($H_{a0} > Y_{fs,0}$ with gas at the junction); the undriven regime retains the dispersed-bubble drag path. In Campaign 2's venting-without-eruption runs the driven-path feed produces the climb in surge pulses (fastest sustained climbs 0.7–1.2 m/s) where the measurements show a steady Taylor-scale ascent (0.42–0.48 m/s); a vertical-annular entrainment closure conditioned on the local gas flux would unify the two paths and smooth the pulsing.

Crown-cavity celerity and the arrival/venting trade-off. Two elements set the timing of the gas-phase sequence. First, a discrete bias in the front-tracking hand-off was identified during this work and removed: releasing the cavity nose cell at 95% of the region-mean layer depth let the detected front advance on 95% of its volume budget, biasing the transit to 1.09 times the nominal celerity. Second, the cavity celerity coefficient itself: Benjamin's $0.542\sqrt{gD}$ is the energy-conserving upper bound, while real (dissipative) cavities run below it, and the middle-pipe transits of the three experimental repetitions correspond to $0.48–0.52\,\sqrt{gD}$. The frozen configuration uses the dissipative value 0.48 (one value for both tests), which places the arrival events inside the repetition scatter (Test A liftoff 7.5 against onsets 7.3–7.9; Test B entry -0.13, eruption -0.20). The trade-off is stated openly: the same crown current that carries the cavity nose also feeds the vent and the narrow-tower pocket-fed core, so the reduced celerity delays the venting-stage events (head-decay onset and interface breakthrough late by 0.4–0.6 units in both tests) and halves the computed narrow-tower climb rate ($V^*_{int} = 0.71$ against 1.43). A closure that separates the nose celerity from the vent feed rate—the former set by the front dynamics, the latter by the pocket overpressure—is the identified refinement.

Film-flow pocket topology in the venting branch. In the wide-tower experiment the gas ascends the shaft as a single Taylor pocket that remains pneumatically continuous with the pipe pocket: its wall film is shear-supported, so the pocket pressure follows the weight of the water column standing *above the pocket nose*, the transducer head decays gradually *during* the climb, and the free surface creeps upward by pocket expansion (Figs. 5 and 10 of [2]; their TPA model encodes this topology by construction, as a lumped pocket with a Batchelor-type film-flow closure). In the present field formulation the tower gas transits as a dispersed phase interleaved with liquid, the tower base carries the weight of the full standing column, and the computed head therefore holds its plateau until breakthrough (Fig. 5); the climb-phase free-surface creep is likewise absent ($V_{fs}^* \approx 0$ against the measured 0.048). We tested a connected-pocket closure that relaxes the pipe-pocket pressure toward the hydrostatic environment of the tower gas nose while crediting the transferred volume to the tower (retained as a solver option). It reproduces the free-surface creep quantitatively (Y_{fs}^* from 0.59 to 0.64 during the climb, against 0.55–0.65 measured) without degrading the validated event times, but the head history develops a spurious hump near the catch instead of the measured monotone decay, and more aggressive relaxation destabilizes the junction dynamics. We conclude that the during-climb head decay is a property of the coherent-pocket topology that a dispersed one-dimensional two-fluid closure cannot represent stably at this grid scale—consistent with [2]'s own choice of a lumped-pocket formulation for exactly this regime—and the frozen configuration retains the plateau-then-collapse behaviour, which preserves the plateau level, the collapse onset time, and the interface kinematics.

Release oscillation damping. The underdamped release-piston oscillation visible in both tests is a genuine mode of the system—a water column on a gas spring—and is annotated as such in the experimental schematic of [2]; the experimental traces damp within about one time unit, the model within several. The damping in the experiment is provided by three-dimensional mixing losses at the valve and coupling, which the constant valve and junction loss coefficients capture in the mean but not cycle-resolved. In the narrow tower the surviving oscillation also lets the column splash briefly to the rim during the release swing before it settles at the quasi-static stand; in Campaign 2 the same simplification sharpens the B-H1 eruption jet (1.71 against 0.92 m/s as event speeds).

Volume-neutral mouth exchange in the reservoir-fed layout. The inverted-glug exchange at the tee mouth—gas up, an equal water volume down—is what keeps the pocket pressure intact while gas transits a standing column, and it is essential to the B-H1 geyser: without it the narrow-riser pocket cannot rebuild the overpressure whose release ejects the column. But in the reservoir-fed layout of Campaign 2 the same exchange cancels part of the one-to-one column lift that the entering gas volume should produce, and the cancellation grows with riser width: the computed B-H6 free-surface rise is +0.12 m against the measured +0.63 m, and across the 63-configuration sweep the under-lift accounts for the one-sided transition-band misses (Section 3.2.3). Gating the exchange off for reservoir-backed cases was tested and rejected: it restores the wide-riser lift but opens the narrow pocket's volume during the balanced venting phase, its pressure never rebuilds, and the B-H1 geyser is lost. The two branches place conflicting demands on a single mouth closure; a formulation that meters the exchange by the local counter-current capacity of the mouth (rather than volume neutrality) is the identified refinement, together with the entrainment closure above.

4.3. Scope of the present validation

The completed campaigns cover three apparatuses and three kinds of evidence: time-resolved two-regime histories in a single-shaft geometry (Campaign 1), regime classification over a systematic parameter sweep in a riser–constant-head geometry (Campaign 2), and a prototype-motivated junction-chamber geometry with a downstream-boundary branch pair and a transient mass-oscillation geyser (Campaign 3), all under one frozen solver configuration and a fully synchronous junction coupling. Within this scope the conclusions are: in Campaign 1, regime selection is exact and both branches are quantitative in their pressure and interface histories, with absolute event times uniformly early by 0.5–0.7 time units; in Campaign 2, the regime boundary is reproduced in structure (both end members, the interaction sense, the branch differentiation of the pipe-end pressure) with a one-sided conservative bias in the transition band (5/7 of Series B, 39/63 of the sweep, no false positives); in Campaign 3, the A2/B3 pair flips branch with downstream boundary alone, C9 phase 1 matches the first transient geyser in timing and first-peak amplitude against both measurement and Eq. (12), while compressive spikes and pocket-driven late geysers remain limited by the frozen elastic celerity and the reduced one-dimensional chamber treatment. Three caveats bound the claims. First, the narrow-shaft closure elements of Test B (crown-datum base coupling, pocket-fed core, flooding-limited entry) are engaged only in the driven regime; their consolidation with the wide-tower configuration into a single junction closure is pending. Second, the volume-neutral mouth exchange places conflicting demands on the two Campaign-2 branches (see above), and its resolution will move the transition-band classifications. Third, Campaign 3 uses a composite one-dimensional chamber representation and does not resolve pocket transport to the junction or chamber gas venting; comparison at larger scale [4] remains a separate target.

5. Conclusions

This paper applied the decoupled two-fluid cut-cell finite-volume framework of [12] to laboratory geysering experiments, representing the vertical shaft as the same one-dimensional two-fluid system rotated to the vertical and coupled to the horizontal conduit through junction-loss closures. Three campaigns were completed, all computed from the experimental initial conditions without prescribed release rates or per-case fitting.

Against the ventilation-shaft experiments of Vasconcelos and Wright [2], the model selected the correct regime in both the non-geysering and geysering configurations. In the non-geysering branch it reproduced the pressure plateau level and duration, the onset of the terminal venting decay, the maximum free-surface rise, and the air–water interface climb velocity. In the geysering branch it reproduced the pre-eruption quasi-static stand of the tower column ($0.84L$ against $0.82L$), the crown-datum pressure plateau (0.76 against 0.76), the eruption to the shaft rim observed in every repetition, the interface and free-surface climb velocities ($V^*_{int} = 1.19$ against 1.43; $V^*_{fs} = 0.42$ against 0.44), and the sharp plateau–collapse of the head at pocket venting; the narrow-shaft closure combines a crown-datum base coupling, a pocket-fed gas core, and a Wallis flooding limit on the entry flux, using geometric and literature-scale constants. The remaining biases are a uniform early shift of the gas-phase event times (0.3–0.6 time units, within the uncertainty of the hand-operated valve opening) and an underdamped release oscillation.

Against the horizontal-pipe/vertical-riser experiments of Cong, Chan and Lee [3], computed with the same frozen solver under the fully synchronous coupling, the model reproduced the branch structure of the published two-parameter geysering criterion: the geysering end member (B-H1, with its compression-surge drive to 1.71 H_0), the entire no-geysering half-plane $D_r/D > 0.62$ and small-volume region, and the branch differentiation of the pipe-end pressure that only a coupled riser–pipe computation can produce. Its errors are one-sided and conservative: two intermediate Series-B runs and 24 of 63 sweep configurations that geyser in the experiment are computed as strong but sub-rim column lifts, with no false positives; the bias is traced to the volume-neutral mouth exchange, whose conflicting demands in the two branches are documented as the primary closure target.

Against the junction-chamber experiments of Liu, Shao and Zhu [5], computed with the same frozen solver in a composite one-dimensional representation, the model reproduced the downstream-boundary branch flip between Cases A2 and B3, placed the B3 eruption on the experimental h–$P_{\max}$ regression line (with acoustically under-resolved peaks at $a_{wh} = 40$ m/s), and matched the first transient geyser of Case C9 in peak timing and magnitude against both measurement and the authors' Eq. (12). Pocket-driven geysers after pocket arrival at the chamber were declared out of scope.

These results support the use of gas-conserving one-dimensional two-fluid models for system-scale geysering screening, where the primary question is regime selection and venting behaviour rather than jet detail, and they identify the junction and entrainment closures as the targets for quantitative improvement in the geysering branch.

Data Availability Statement

The frozen solver snapshots, digitized experimental curves, and comparison scripts used to produce the results in Section 3 are available from the corresponding author on request.

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